

2 Differentiation

3 Differentiation II

Differentiation



Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 3A03 Real Analysis I

Instructor: David Earn

Lecture 2
Differentiation
Wednesday 8 January 2025

Survey to do right now

- Please go to
https://www.childsmath.ca/childsa/forms/main_login.php
- Click on **Math 3A03**
- Click on **Take Class Poll**
- Fill in poll **Survey 2**
- .

Announcements

- Results of Survey 1
- Results of Survey 2

Background / reminder

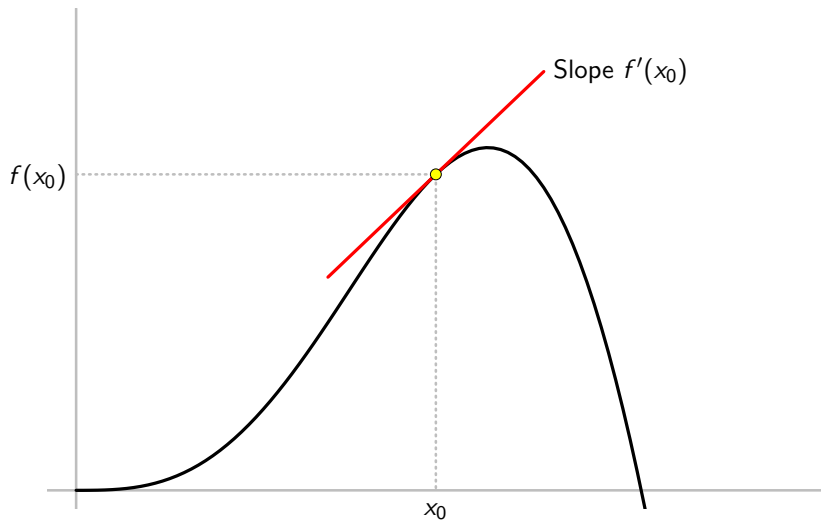
Definition (Cauchy sequence)

A sequence $\{s_n\}$ is said to be a *Cauchy sequence* iff for all $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that if $m \geq N$ and $n \geq N$ then $|s_n - s_m| < \varepsilon$.

Poll: another background check

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- .

The Derivative



The Derivative

Definition (Derivative)

Let f be defined on an interval I and let $x_0 \in I$. The **derivative** of f at x_0 , denoted by $f'(x_0)$, is defined as

$$f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0},$$

provided either that this limit exists or is infinite. If $f'(x_0)$ is finite we say that f is **differentiable** at x_0 . If f is differentiable at every point of a set $E \subseteq I$, we say that f is differentiable on E . If E is all of I , we simply say that f is a **differentiable function**.

Note: “Differentiable” and “a derivative exists” always mean that the derivative is finite.

The Derivative

Example

$f(x) = x^2$. Find $f'(2)$.

$$f'(2) = \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} \frac{(x + 2)(x - 2)}{x - 2} = \lim_{x \rightarrow 2} x + 2 = 4$$

Note:

- In the first two limits, we must have $x \neq 2$.
- But in the third limit, we just plug in $x = 2$.
- Two things are equal, but in one $x \neq 2$ and in the other $x = 2$.
- Good illustration of why it is important to define the meaning of limits rigorously.

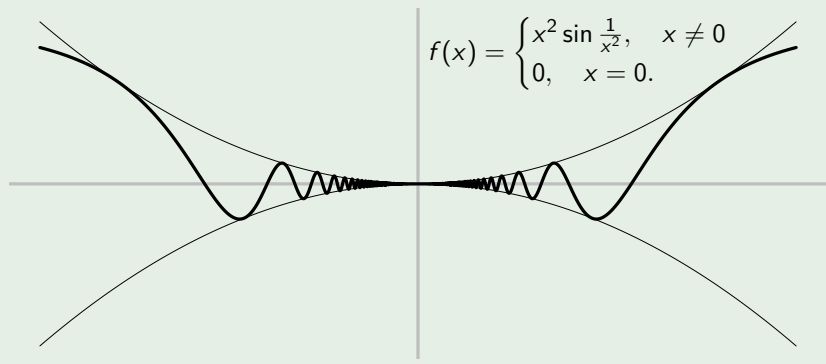
Poll

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- Click on Math 3A03
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- Fill in poll **Derivatives: Differentiable at 0**
- .

The Derivative

Example

Let f be defined in a neighbourhood I of 0, and suppose $|f(x)| \leq x^2$ for all $x \in I$. Is f necessarily differentiable at 0? e.g.,



The Derivative

Example (Trapping principle)

Suppose $f(x) = \begin{cases} x^2 \sin \frac{1}{x^2}, & x \neq 0 \\ 0, & x = 0. \end{cases}$ Then:

$$\forall x \neq 0 : \left| \frac{f(x) - f(0)}{x - 0} \right| = \left| \frac{f(x)}{x} \right| = \left| \frac{x^2 \sin \frac{1}{x^2}}{x} \right| = \left| x \sin \frac{1}{x^2} \right| \leq |x|$$

Therefore:

$$|f'(0)| = \left| \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} \right| = \lim_{x \rightarrow 0} \left| \frac{f(x) - f(0)}{x - 0} \right| \leq \lim_{x \rightarrow 0} |x| = 0.$$

$\therefore f$ is differentiable at 0 and $f'(0) = 0$. □

The Derivative

Definition (One-sided derivatives)

Let f be defined on an interval I and let $x_0 \in I$. The **right-hand derivative** of f at x_0 , denoted by $f'_+(x_0)$, is the limit

$$f'_+(x_0) = \lim_{x \rightarrow x_0^+} \frac{f(x) - f(x_0)}{x - x_0},$$

provided either that this one-sided limit exists or is infinite.

Similarly, the **left-hand derivative** of f at x_0 , denoted by $f'_-(x_0)$, is the limit

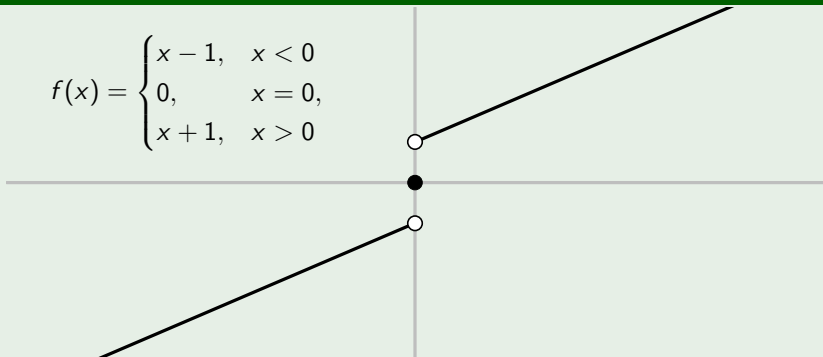
$$f'_-(x_0) = \lim_{x \rightarrow x_0^-} \frac{f(x) - f(x_0)}{x - x_0}.$$

Note: If x_0 is not an endpoint of the interval I then f is differentiable at x_0 iff $f'_+(x_0) = f'_-(x_0) \neq \pm\infty$.

The Derivative

Example

$$f(x) = \begin{cases} x - 1, & x < 0 \\ 0, & x = 0, \\ x + 1, & x > 0 \end{cases}$$



- Same slope from left and right. Why isn't f differentiable???
- $\lim_{x \rightarrow 0^-} f'(x) = \lim_{x \rightarrow 0^+} f'(x) = \lim_{x \rightarrow 0} f'(x) = 1.$
- $f'_-(0) = f'_+(0) = f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \infty.$

The Derivative

- Higher derivatives: we write
 - $f'' = (f')'$ if f' is differentiable;
 - $f^{(n+1)} = (f^{(n)})'$ if $f^{(n)}$ is differentiable.
- Other standard notation for derivatives:

$$\frac{df}{dx} = f'(x)$$

$$D = \frac{d}{dx}$$

$$D^n f(x) = \frac{d^n f}{dx^n} = f^{(n)}(x)$$

REMINDER: Algebra of limits

Theorem (Algebraic operations on limits of sequences)

Suppose $\{s_n\}$ and $\{t_n\}$ are *convergent sequences* and $C \in \mathbb{R}$.

1 $\lim_{n \rightarrow \infty} C s_n = C \left(\lim_{n \rightarrow \infty} s_n \right) ;$

2 $\lim_{n \rightarrow \infty} (s_n + t_n) = \left(\lim_{n \rightarrow \infty} s_n \right) + \left(\lim_{n \rightarrow \infty} t_n \right) ;$

3 $\lim_{n \rightarrow \infty} (s_n - t_n) = \left(\lim_{n \rightarrow \infty} s_n \right) - \left(\lim_{n \rightarrow \infty} t_n \right) ;$

4 $\lim_{n \rightarrow \infty} (s_n t_n) = \left(\lim_{n \rightarrow \infty} s_n \right) \left(\lim_{n \rightarrow \infty} t_n \right) ;$

5 if $t_n \neq 0$ for all n and $\lim_{n \rightarrow \infty} t_n \neq 0$ then

$$\lim_{n \rightarrow \infty} \left(\frac{s_n}{t_n} \right) = \frac{\lim_{n \rightarrow \infty} s_n}{\lim_{n \rightarrow \infty} t_n} .$$

(TBB §2.7, and problem 2.7.4)

REMINDER: Algebra of limits

Theorem (Algebraic operations on limits of functions)

Suppose $f, g : \mathbb{R} \rightarrow \mathbb{R}$, $x_0 \in \mathbb{R}$, the limits as $x \rightarrow x_0$ of $f(x)$ and $g(x)$ both exist, and $C \in \mathbb{R}$.

$$1 \quad \lim_{x \rightarrow x_0} C f(x) = C \left(\lim_{x \rightarrow x_0} f(x) \right) ;$$

$$2 \quad \lim_{x \rightarrow x_0} (f(x) + g(x)) = \left(\lim_{x \rightarrow x_0} f(x) \right) + \left(\lim_{x \rightarrow x_0} g(x) \right) ;$$

$$3 \quad \lim_{x \rightarrow x_0} (f(x) - g(x)) = \left(\lim_{x \rightarrow x_0} f(x) \right) - \left(\lim_{x \rightarrow x_0} g(x) \right) ;$$

$$4 \quad \lim_{x \rightarrow x_0} (f(x)g(x)) = \left(\lim_{x \rightarrow x_0} f(x) \right) \left(\lim_{x \rightarrow x_0} g(x) \right) ;$$

$$5 \quad \text{if } g(x) \neq 0 \text{ for } x \in (x_0 - \delta, x_0 + \delta) \text{ for some } \delta > 0, \text{ and} \\ \lim_{x \rightarrow x_0} g(x) \neq 0 \text{ then } \lim_{x \rightarrow x_0} \left(\frac{f(x)}{g(x)} \right) = \frac{\lim_{x \rightarrow x_0} f(x)}{\lim_{x \rightarrow x_0} g(x)} .$$

The Derivative

Theorem (Differentiable $\implies$ continuous)

If f is defined in a neighbourhood I of x_0 and f is differentiable at x_0 then f is continuous at x_0 .

Proof.

Must show $\lim_{x \rightarrow x_0} f(x) = f(x_0)$, i.e., $\lim_{x \rightarrow x_0} (f(x) - f(x_0)) = 0$.

$$\begin{aligned}\lim_{x \rightarrow x_0} (f(x) - f(x_0)) &= \lim_{x \rightarrow x_0} \left(\frac{f(x) - f(x_0)}{x - x_0} \times (x - x_0) \right) \\ &= \lim_{x \rightarrow x_0} \left(\frac{f(x) - f(x_0)}{x - x_0} \right) \times \lim_{x \rightarrow x_0} (x - x_0) \\ &= f'(x_0) \times 0 = 0,\end{aligned}$$

where we have used the theorem on the algebra of limits. □



Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 3A03 Real Analysis I

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Lecture 3
Differentiation II
Friday 10 January 2025

Course evaluation will be revised as follows:

- We will not have quizzes
- 5% for participating in at least 80% of in-class polls
- 15% for participating in assignments, based on multiple choice (MC) questions:

$$\text{assignment mark} = \frac{\text{number MC questions answered}}{\text{total number MC questions assigned}}$$

- 30% for midterm test on Thurs 27 Feb 2025
- 50% for final exam in April
- Note: If your final exam mark is better than your midterm test mark then the final exam mark will replace the midterm test mark.
- Important: Do NOT skip the midterm. Even if you don't feel well prepared, write it for practice so you are better prepared for writing the final exam.
- If you must miss the midterm (e.g., illness or accepting a Nobel prize), your final exam mark will replace it.

Tentative plan for assignments

- There will be regular assignments.
- Each question will have a multiple choice component (probably on [childsmath](#)). Only participation counts for marks; you will get the same credit for correct and incorrect answers, or for selecting “I haven’t had time to think about this yet”.
- Optionally, full solutions/proofs can be written up and submitted on [crowdmark](#). Feedback will be given, but no marks. The purpose is to help you prepare better for the test and exam.
- If you’re not sure if your proof is complete, or you got stuck and don’t know how to complete it, make that clear in the document that you submit on [crowdmark](#), so the TA can focus on the help you need.
- Always try your best to solve problems on your own first. But if you used stackexchange or ChatGPT or whatever for help, provide a URL to your source if possible, so it is easier for the TA to provide the help you need.
- Make the best possible use of the TA’s time: say what you think you do or don’t understand.

Last time...

- Definition of the derivative.
 - Example: Trapping Principle
- Defined one-sided derivatives
 - Example
- Proved differentiable $\implies$ continuous.

More on the derivative

Theorem (Algebra of derivatives)

Suppose f and g are defined on an interval I and $x_0 \in I$. If f and g are differentiable at x_0 then $f + g$ and fg are differentiable at x_0 . If, in addition, $g(x_0) \neq 0$ then f/g is differentiable at x_0 . Under these conditions:

1 $(cf)'(x_0) = cf'(x_0)$ for all $c \in \mathbb{R}$;

2 $(f + g)'(x_0) = (f' + g')(x_0)$;

3 $(fg)'(x_0) = (f'g + fg')(x_0)$;

4 $\left(\frac{f}{g}\right)'(x_0) = \left(\frac{gf' - fg'}{g^2}\right)(x_0) \quad (g(x_0) \neq 0).$

(TBB Theorem 7.7, p. 408)

The Derivative

Theorem (Chain rule)

Suppose f is defined in a neighbourhood U of x_0 and g is defined in a neighbourhood V of $f(x_0)$ such that $f(U) \subseteq V$. If f is differentiable at x_0 and g is differentiable at $f(x_0)$ then the composite function $h = g \circ f$ is differentiable at x_0 and

$$h'(x_0) = (g \circ f)'(x_0) = g'(f(x_0))f'(x_0).$$

Informally, if $y = f(x)$ and $z = g(y)$ then $\frac{dz}{dx} = \frac{dz}{dy} \frac{dy}{dx}$.

(TBB §7.3.2, p. 411)

Why the chain rule is plausible

The derivative of $g \circ f$ at x_0 is the limit as $x \rightarrow x_0$ of the difference quotient

$$\frac{g(f(x)) - g(f(x_0))}{x - x_0} = \frac{g(f(x)) - g(f(x_0))}{f(x) - f(x_0)} \cdot \frac{f(x) - f(x_0)}{x - x_0} \quad (\spadesuit)$$

Recall: $f'(x_0)$ exists $\implies f$ continuous at x_0
 $\implies f(x) \rightarrow f(x_0)$ as $x \rightarrow x_0$.

Can we take the limit as $x \rightarrow x_0$ and conclude that $(g \circ f)'(x_0) = g'(f(x_0)) \cdot f'(x_0)$?

- What if $f(x) = 0$ for all x ?
- What if f is a constant function?
- What if $f(x) = f(x_0)$ for some $x \neq x_0$?
- Can we use $(\spadesuit)$ to prove the chain rule?

Poll

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- Click on **Math 3A03**
- Click on **Take Class Poll**
- Fill in poll **Derivatives: Chain Rule**
- .

REMINDER: limits of functions

Theorem (Equivalence of ε - δ and sequence definitions of limits)

Let $a < x_0 < b$, $I = (a, b)$, and $f : I \setminus \{x_0\} \rightarrow \mathbb{R}$. Then the following two definitions of

$$\lim_{x \rightarrow x_0} f(x) = L$$

are equivalent:

- 1 for all $\varepsilon > 0$ there exists $\delta > 0$ such that if $0 < |x - x_0| < \delta$ then $|f(x) - L| < \varepsilon$.
- 2 for every sequence $\{x_n\}$ of points in $I \setminus \{x_0\}$,

$$\lim_{n \rightarrow \infty} x_n = x_0 \implies \lim_{n \rightarrow \infty} f(x_n) = L.$$

Note: The deleted neighbourhood $(I \setminus \{x_0\})$ can be replaced by any set on which f is defined and x_0 is an accumulation point.

Proof of the chain rule.

- 1 Suppose there is an open interval I , with $x_0 \in I$, and $f(x) \neq f(x_0)$ for all $x \in I \setminus \{x_0\}$. Then we can take the limit $x \rightarrow x_0$ in  and we get the [chain rule](#).
- 2 Next suppose that no open interval like the one hypothesized above exists. Then, in any open interval containing x_0 , there must be at least one point $x \neq x_0$ for which $f(x) = f(x_0)$. Therefore, we can construct a sequence of open intervals I_n , with lengths decreasing to 0, such that each I_n contains x_0 and a point $x_n \neq x_0$ with $f(x_n) = f(x_0)$. Therefore, since $f'(x_0)$ exists, and we recall the [previous slide](#), we can compute $f'(x_0)$ via

$$f'(x_0) = \lim_{n \rightarrow \infty} \frac{f(x_n) - f(x_0)}{x_n - x_0} = \lim_{n \rightarrow \infty} \frac{0}{x_n - x_0} = 0.$$

We can also show that $(g \circ f)'(x_0) = 0$, using the sequence definition on the [previous slide](#). *Try to fill in this last detail*, or look it up (TBB [§7.3.2, p. 411](#)).

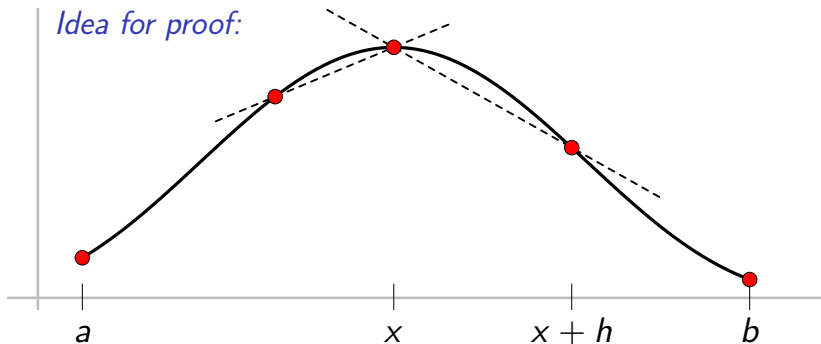
Note: TBB's proof leaves out the proof that $f'(x_0) = 0$ in case 2 above. 

More on the derivative

Theorem (Derivative at local extrema)

Let $f : (a, b) \rightarrow \mathbb{R}$. If x is a maximum or minimum point of f in (a, b) , and f is differentiable at x , then $f'(x) = 0$.

Note: f need not be differentiable or even continuous at other points.



More on the derivative

Proof that the derivative vanishes at local extrema.

If f has a local maximum at $x \in (a, b)$, then for sufficiently small $h > 0$ we must have

$$\frac{f(x+h) - f(x)}{h} \leq 0 \leq \frac{f(x) - f(x-h)}{h}$$

Since f is differentiable at x , it is left and right differentiable at x , so we can evaluate the limits as $h \rightarrow 0$ to obtain

$$f'_+(x) \leq 0 \leq f'_-(x).$$

But since f is differentiable at x , the left and right derivatives must be equal, hence $f'(x) = 0$. □

The Mean Value Theorem

Theorem (Rolle's theorem)

If f is continuous on $[a, b]$ and differentiable on (a, b) , and $f(a) = f(b)$, then there exists $x \in (a, b)$ such that $f'(x) = 0$.

Proof.

f continuous on $[a, b] \implies f$ has a max and min value on $[a, b]$. If either a max or min occurs at $x \in (a, b)$ then $f'(x) = 0$. If no max or min occurs in (a, b) then they must both occur at the endpoints, a and b . But $f(a) = f(b)$, so f is constant. Hence $f'(x) = 0 \forall x \in (a, b)$. $\square$

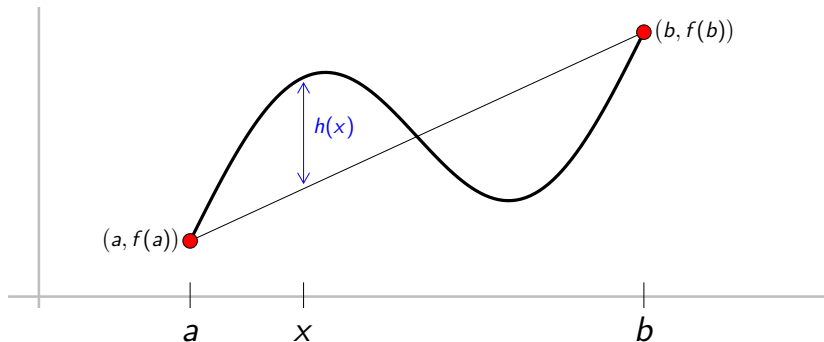
Theorem (Mean value theorem)

If f is continuous on $[a, b]$ and differentiable on (a, b) then there exists $x \in (a, b)$ such that

$$f'(x) = \frac{f(b) - f(a)}{b - a}.$$

The Mean Value Theorem

Idea for proof:



Proof.

Apply [Rolle's theorem](#) to

$$h(x) = f(x) - \left[f(a) + \left(\frac{f(b) - f(a)}{b - a} \right) (x - a) \right].$$



The Mean Value Theorem

Example

$f'(x) > 0$ on an interval $I \implies f$ strictly increasing on I .

Proof:

Suppose $x_1, x_2 \in I$ and $x_1 < x_2$. We must show $f(x_1) < f(x_2)$.

Since $f'(x)$ exists for all $x \in I$, f is certainly differentiable on the closed subinterval $[x_1, x_2]$.

Hence by the [Mean Value Theorem](#) $\exists x_* \in (x_1, x_2)$ such that

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = f'(x_*).$$

But $x_2 - x_1 > 0$ and since $x_* \in I$, we know $f'(x_*) > 0$.

$\therefore f(x_2) - f(x_1) > 0$, *i.e.*, $f(x_1) < f(x_2)$. □